COURSE SYLLABUS & SETUP GUIDE · COURSE 4S4TWS_21

Agentic AI Engineering for Responsible Business

A 16-hour practical engineering course on building tool-using agents, grounded document search, and multi-agent workflows with Google GenAI & Google ADK.

INSTRUCTOR	PROGRAM & COHORT	SCHEDULE
Chouaieb Nemri	**Year 4 · BSc Data Science**	**Wed Oct 14 & Wed Oct 21, 2026 (2 × 8h)**

1. What This Course Is About

Over two full Wednesdays (October 14 and October 21, 2026), we will move past simple chatbot prompting and build production-grade software agents in Python. In sustainability, supply chain auditing, and corporate finance, a model that guesses numbers or invents legal articles is a liability. You will learn how to wrap LLMs inside deterministic Python control loops that look up verified supplier records, compute exact EU carbon taxes in Python, cross-check marketing claims against EU regulations, and enforce safety guardrails before taking action.

> **What You Will Walk Away With**
>
> By the end of Day 2, you will have built four hands-on lab notebooks (`TP1` – `TP4`) and a complete **Responsible Supply Chain Auditor** (`04_Homework_Oct14_to_Oct21/`) that you can showcase in technical interviews or adapt for your Year 4 capstone project.

2. Schedule & Format (16 Hours — Remote via Google Meet)

All sessions take place live via Google Meet (`08:00–12:00` and `14:15–18:15 CET`). Every two-hour block combines architectural concepts, live instructor coding, and hands-on pod sprints:

Session	Date & Time	Topics Covered	Hands-On Work
`014` (AM)	Wed Oct 14 `08:00–12:00`	Why prompts fail in production, the ReAct loop (`Think -> Act -> Observe`), model routing, Pydantic JSON schemas, and Python tool calling.	**TP1**: Scope 3 Supplier ESG & Carbon Tax Agent
`015` (PM)	Wed Oct 14 `14:15–18:15`	Stateful sessions with the Interactions API, vector search with `gemini-embedding-2` , and citation grounding.	**TP2**: EU Greenwashing Claims Auditor + Homework Kickoff
`016` (AM)	Wed Oct 21 `08:00–12:00`	Homework debrief, multi-agent decomposition, and Google ADK (`Agent` , `SequentialAgent` , `ParallelAgent` , `LoopAgent`).	**TP3**: Multi-Agent Green CapEx Committee
`017` (PM)	Wed Oct 21 `14:15–18:15`	Automated trajectory evaluation, indirect prompt injection (`SUP-010`) defense, and Python callback guardrails.	**TP4**: Safety Guardrails + Live Pod Demos

3. Model Lineup: Lecture Concepts vs. Free-Tier Hands-On Execution

In the lectures, we study Google's frontier **Gemini 4 Argon & Gemini 3.8 Flash** architecture. For all hands-on Labs (`TP1` – `TP4`) and the Homework, we run on Google's open-weights **Gemma 4** models via the official `google-genai` SDK so every student gets **generous 100% Free-Tier quotas** on their personal Google AI Studio key:

- **`gemma-4-26b-a4b-it` (`MODEL_FAST` — MoE Workhorse, 4B Active Params)**: Default model for `TP1` – `TP4` and the Homework. Natively supports `system_instruction` , Pydantic `response_schema` , parallel function calling, and Google ADK (`1,500 RPD` & `16,000 TPM` free per student).
- **`gemma-4-31b-it` (`MODEL_FRONTIER` — Dense 31B Reasoning & Committee Chair)**: Used for complex synthesis (`CommitteeChair` in `TP3`) and model routing comparisons, with its own independent `1,500 RPD` & `16,000 TPM` Free-Tier quota bucket.
- **`gemini-embedding-2` (Semantic Vector Search)**: Embeds EU regulatory articles and corporate claims for grounded RAG in `TP2` and the Homework.

4. Student Pods & Collaboration

Our 12-student cohort is organized into four 3-person pods for breakout lab sprints and the Day 2 Capstone Defense:

Pod	Student Name	emlyon Email
Pod Alpha	Mme Jiani XU	`jiani.xu@edu.em-lyon.com`
Pod Alpha	Mme Yuxin GONG	`yuxin.gong@edu.em-lyon.com`
Pod Alpha	M. Simon GRANDIDIER	`simon.grandidier@edu.em-lyon.com`
Pod Beta	Mme Mélissa CEHIC	`melissa.cehic@edu.em-lyon.com`
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Pod Gamma	Mme Charlotte DEMODE	`charlotte.demode@edu.em-lyon.com`
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Pod Delta	M. Harshavardhan Reddy MALIKIREDDY	`harshavardhan.malikireddy@edu.em-lyon.com`
Pod Delta	Mme Naomi HELMBOLD	`naomi.helmbold@edu.em-lyon.com`

5. Grading Breakdown

Component	Weight	Format	What We Look For
In-Class Labs (`TP1` – `TP4`)	**40%** (10% each)	Individual / Pod	Starter notebook (`_Starter_*.ipynb`) runs cleanly end-to-end and passes verification checks.
Graded Homework (`Oct 14 -> Oct 20`)	**35%**	Individual	Passing `check_milestone_1..4` and the 10-case auto-grader `grade_my_agent()` in `agent_utils.py` .
Day 2 Pod Capstone Defense	**25%**	Pod (3 students)	15-minute live defense (`16:45–17:55` on Oct 21) + surviving a live challenger query from another pod.

6. Quick Setup & GitHub Workflow Before Wednesday, Oct 14 (5 Minutes)

The course GitHub repository publishes the **Starter notebooks (`énoncés`)**, datasets, and `agent_utils.py` on `main` . Official reference solutions (`_Solution.ipynb`) are published via **GitHub Releases** after Day 2 concludes.

```
# 1. Clone the course repository and create a Python 3.11+ virtual environment
git clone https://github.com/cnemri/ecl-agentic-ai-course.git && cd ecl-agentic-ai-course
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt

# 2. Export your personal Free-Tier Google AI Studio API Key (https://aistudio.google.com/apikey)
export GEMINI_API_KEY="your-personal-ai-studio-key"
```

> **Free-Tier Rate Limit Protection & Offline Fallback Built In**
>
> Every starter notebook and `agent_utils.py` include `strip_thought_parts()` (cutting Gemma 4 multi-turn input tokens by 65%), automatic `429` cooldown retries, and a deterministic offline fallback if your Wi-Fi drops.